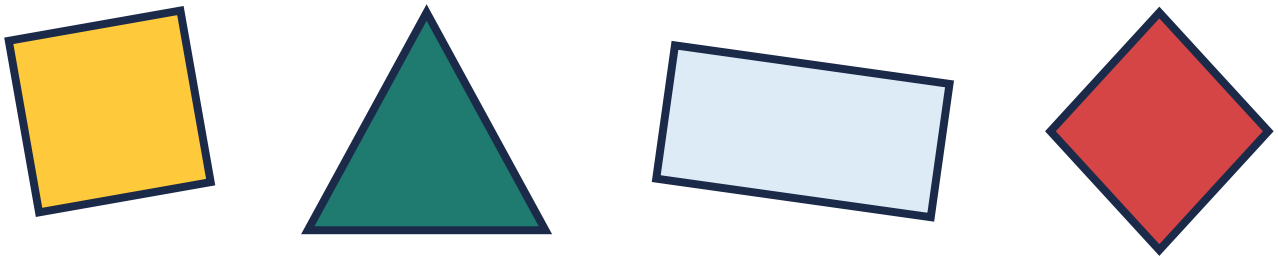
FOR PARENTS • LEVEL 6, AFTER THE PSLE

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Paper, a pencil and, for coding lessons, a computer with Scratch (free at scratch.mit.edu). Let your child show you the program and explain what each block does.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Go further questions if your child enjoys them.

The three levels

Start here is for everyone. **Think harder** stretches the idea. **Go further** and the **Investigation** are for when your child wants more, and it is fine to leave them for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Code the Pattern

LESSON 1 • SEQUENCES AND THE NTH TERM

Start here

1 Write the next two terms: 8, 13, 18, 23, ...

2 Matchstick triangles in a row. Pattern 1 is 1 triangle (3 sticks). Pattern 2 is 2 triangles side by side (5 sticks). Pattern 3 is 3 triangles (7 sticks). Use matchsticks or draw them. How many sticks does pattern 4 need?

3 Write out the terms one by one to find the 10th term of 7, 10, 13, 16, ...

4 Write the next two terms: 40, 36, 32, 28, ...

Think harder

5 Find the 100th term of 12, 19, 26, 33, ...

6 Is 200 a term of 2, 9, 16, 23, ...? Explain.

7 Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?

What your child learns

Finding the rule of a number pattern, testing it in a short program, and finding the 100th term without listing them all.

Tip for you

Make a matchstick pattern of triangles or squares. Ask how many sticks the 10th figure needs before you build it.

Ask

"Can you test your rule on the next figure?"

Go further

8 Pattern 1 is 1 tile, pattern 2 is a 2 by 2 square of tiles, pattern 3 is a 3 by 3 square. How many tiles are in pattern 10? How many more tiles does pattern 11 need than pattern 10?

9 Ben counts 6, 11, 16, 21, ... and Mei counts 83, 81, 79, 77, ... They say their terms at the same time. When do they say the same number, and what is it?

W1 Prime Hunters

LESSON 2 • PRIMES, FACTORS AND PRIME FACTORISATION

Start here

1 List all the factors of 18. Hint: find pairs of numbers that multiply to 18.

2 A number is called prime if it has exactly two factors: 1 and itself. For example, 7 is prime (factors 1 and 7) but 8 is not (factors 1, 2, 4, 8). Is 21 prime? Is 23 prime? List the factors of each.

3 29 sweets are shared equally among 4 children. How many sweets are left over? (In code, the left-over number is written 29 mod 4.)

4 Write 30 as a multiplication of as many numbers (bigger than 1) as you can. For example, 30 = 5 × 6 uses two numbers. Can you use three? Can you use four?

Think harder

5 How many factors does 64 have? List them.

6 Find the smallest number that has exactly 6 factors.

What your child learns

Prime numbers, factors, factor trees, and a program that counts factors.

Tip for you

Pick a number under 100 and ask whether it is prime. Let your child test it one divisor at a time.

Ask

"What would you try next if it is not divisible by 2, 3 or 5?"

7 Write 100 as the sum of two primes in every possible way.

Go further

8 What is the largest prime number below 200?

9 Ali multiplies two different primes and gets 91. Mei multiplies three different primes and gets 105. Which primes did each of them use?

W2 Below Zero

LESSON 3 • NEGATIVE NUMBERS

Start here

1 On a thermometer, 0 °C is where water freezes. Colder than that, numbers have a minus sign: −3 °C means 3 degrees below zero. Draw a thermometer or number line and write these in order, coldest first: −4, 2, −10, 0, −1.

2 It is −3 °C (3 degrees below zero). The temperature rises by 8 degrees. What is it now? Count up on a thermometer drawing.

3 It is 2 °C. The temperature falls by 7 degrees. What is it now? Count down on a thermometer drawing: 2, 1, 0, then below zero.

4 Which is colder, −6 °C or −11 °C? By how many degrees? Mark both on a thermometer drawing.

Think harder

5 A sprite starts at x = a. It runs “change x by −8” and then “change x by 5”. It ends at x = −1. What was a? Work backwards from the end, and check by running it forwards.

What your child learns

Negative numbers on a number line, temperatures and floors below zero, and programs that go up and down.

Tip for you

Look at the weather in a cold city and ask for the difference between its low and ours.

Ask

"Which way do you move on the number line?"

6 A diver is at −14 m. A seagull is flying at 9 m. How far apart are they?

7 Use the numbers −2, 3 and 5 once each to fill the gaps in ____ − ____ − ____. What is the biggest answer you can make? What is the smallest?

Go further

8 Put −4, −3, −2, −1, 0, 1, 2, 3, 4 in a 3 by 3 grid so that every row, every column and both diagonals add up to 0.

9 At midnight it was −6 °C. It rose 2 degrees every hour until 7 a.m., then 3 degrees every hour. What was the temperature at 10 a.m.?

W2 Algebra Machines

LESSON 4 • EXPRESSIONS AND EQUATIONS

Start here

1 A number machine multiplies IN by 4 and then adds 1. (We write this OUT = 4n + 1.) What comes OUT when 6 goes IN? Try 2 and 3 first.

2 A machine changes IN 1, 2, 3, 4 into OUT 3, 5, 7, 9. Look at how OUT goes up each time. What does the machine do? What will come OUT when 10 goes IN?

3 I think of a number and multiply it by 3. I get 54. What was my number? (Guess and check, or share 54 into 3 equal groups.)

4 A machine multiplies by 6, then takes away 4. OUT is 38. What was IN? Try some numbers, or undo the steps.

Think harder

5 IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?

6 Solve 9n − 7 = 65.

What your child learns

Number machines and simple algebra: finding the rule, working backwards, and a program that repeats until it hits a target.

Tip for you

Say a number-machine riddle: "I think of a number, double it, add 3 and get 17." Swap and let your child make one up.

Ask

"What is the opposite of each step?"

7 Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.

Go further

8 A machine does × 3, then − 4. Which IN number comes out as the SAME number?

9 Two machines are joined: the OUT of the first goes into the second. Each one does × 2, then + 1. IN 5 gives OUT 23. Which IN gives OUT 63?

W3 LESSON 5 • COUNTING Every Possible Way

Start here

1 A shop sells 3 shirts and 5 pairs of trousers. How many different outfits (1 shirt and 1 pair of trousers) can you make? Draw lines to match each shirt with each pair of trousers.

2 List every 2-digit number you can make from the cards 4, 6 and 9, using each card at most once.

3 Four friends meet. Every pair of friends shakes hands once. How many handshakes are there? Act it out, or draw 4 dots and join every pair.

4 How many times does this say “hi”? repeat 4 { repeat 6 { say hi } } (The inner repeat runs all 6 times for each time round the outer repeat.)

Think harder

5 A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?

What your child learns

Counting every possible way: tree diagrams, the multiplication rule, handshakes, and nested loops.

Tip for you

Ask how many outfits can be made from 3 tops and 4 skirts, then draw all of them.

Ask

"How do you know you have not missed or counted twice?"

6 How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?

7 In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?

Go further

8 How many different triangles can you make by joining 3 of the 5 corners of a pentagon? What about a hexagon?

9 How many 3-digit numbers have three different digits? (Remember: a number cannot start with 0.)

W3 LESSON 6 • PROBABILITY
Fair or Unfair?

Start here

1 Roll a dice. Write down every possible result. How many of them are even? So the chance of an even number is __ out of __. Write it as a fraction.

2 A bag has 3 red and 7 yellow counters. You pick one without looking. How many counters are there altogether? What is the chance (as a fraction) that you pick yellow?

3 You flip a coin 200 times. About how many heads do you expect? Will it be exactly that number?

4 A spinner has 4 equal parts: red, red, blue, green. Write the chance of spinning red as a fraction.

Think harder

5 A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?

6 Roll two dice and add. What is the probability that the total is 9?

What your child learns

Probability as a fraction, fair and unfair games, and a program that rolls a die 1000 times.

Tip for you

Play a dice game with a friend. Ask whether it is fair, and then keep a tally of 50 throws.

Ask

"Why is it close, but not exactly what we expected?"

7 A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?

Go further

8 Three coins are flipped. What is the probability of getting exactly 2 heads?

9 A number from 1 to 30 is picked at random. What is the probability that it is a multiple of 3 or a multiple of 5?

W4 LESSON 7 • ANGLES IN POLYGONS
Turtle Geometry

Start here

1 Walk a square: walk forward, turn on the spot by 90° to the right, and repeat, until you are back at the start facing the way you began. You turn 90° at each of the 4 corners. How many degrees did you turn altogether? (A full turn on the spot is 360°.)

2 A turtle runs: repeat 6 [move 100, turn ↻ 60]. It draws a hexagon. How many degrees does it turn altogether? What do you notice?

3 A turtle draws an equilateral triangle: repeat 3 [move 100, turn ↻ ___]. It turns the same amount each time and ends up facing the way it started (a full turn is 360°). Guess and check, or draw it with a ruler. What number goes in the gap?

4 A turtle repeats [move 100, turn ↻ 72] again and again. How many repeats until it first faces the way it started? Add up the turns, 72, 144, ... until you reach a full turn (360). What shape does it draw?

Think harder

5 Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?

What your child learns

Turtle geometry: turning angles, regular polygons and stars, and total turning.

Tip for you

Walk a regular shape on the floor: five steps, turn, repeat. How much did you turn in total when you came back?

Ask

"Why does the turtle always turn 360 degrees in total?"

6 Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?

7 Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?

Go further

8 Ali wants a regular pentagon. He types the inside angle by mistake: repeat 5 [move 100, turn ↻ 108]. Does his shape close? How many repeats would make it close, and what would it draw?

9 How many different stars with 9 points can a turtle draw using equal moves and equal turns? Give the turns.

W4 Secret Codes

LESSON 8 • CLOCK (MODULAR) ARITHMETIC

Start here

1 It is 10 o'clock. What time will it be 5 hours later?

2 A clock has 12 numbers. Start at 12 and move 20 hours forward. Where do you stop? (This is written 20 mod 12: the remainder when 20 is divided by 12.)

3 What is the remainder when 45 is divided by 7? Hint: count up in 7s.

4 Today is Monday. What day will it be 10 days from today?

Think harder

5 The pattern CODECODECODE... goes on and on. What is the 100th letter?

6 It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?

What your child learns

Secret codes: clock arithmetic, shift codes, cracking a code, and a program that scrambles a message.

Tip for you

Send each other a short message in a shift code and crack it by looking for the most common letters.

Ask

"What happens when you go past Z?"

7 Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.

Go further

8 Mei makes a new code: multiply each letter number by 2, then take mod 26. Why can her friend not always decode it?

9 Ben's code multiplies each letter number by 3, then takes mod 26. Code the word CAB. Do two letters ever get the same code letter?

W5 Data Detectives

LESSON 9 • AVERAGES AND DATA

Start here

1 Find the mean of 3, 5, 7 and 9. Hint: share the total equally, or even out the towers.

2 Find the mean of 10, 0, 8 and 6.

3 The mean of 5 numbers is 11. What is their total? Think: if the total were shared equally between 5 people, each would get 11.

4 Four bags have a mean mass of 2.5 kg. What is their total mass?

Think harder

5 The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?

6 Five whole numbers have a mean of 8, a median of 10 and a mode of 12 (12 appears more often than any other number). Find them. Is there more than one answer?

What your child learns

Data: mean, median and mode, what changes the mean, and a program that finds the mean of a list.

Tip for you

Collect the heights or pocket money of your family and work out the mean and the median. Which describes them better?

Ask

"When is the mean misleading?"

7 Write 5 whole numbers with a mean of 6 and a mode of 4.

Go further

8 Five numbers have a mean of 8. One number is removed and the mean of the other four is 7. (a) Which number was removed? (b) Ali says: “Taking a number out always makes the mean go down.” Is he right? Explain, using an example.

9 A basketball player’s mean score for 9 games is 30 points. (a) How many points must she score in the 10th game to raise her mean to 32? (b) The most she has ever scored in one game is 60. What is the highest mean she could have after game 10?

W5 LESSON 10 • MIXED: MATHS AND CODE
Showcase Challenge

Start here

1 Find the 20th term of 5, 8, 11, 14, … (Write the terms out, or look for a shortcut.)

2 Break 60 into smaller and smaller multiplications until every number is prime (2, 3, 5, 7, 11, …). Write 60 as a product of primes.

3 The temperature is −3 °C (3 degrees below zero). It falls by 5 degrees. What is it now? Count down on a thermometer drawing.

4 A machine does × 4, then − 6. What comes out when 5 goes in?

Think harder

5 A row of squares uses 4, 7, 10, 13, … matchsticks. Which pattern number uses exactly 100 matchsticks?

6 Find all the prime numbers between 80 and 100.

What your child learns

Showcase: a mixed challenge round, then planning, building and testing a small puzzle program.

Tip for you

Ask your child to explain their program to you, block by block. Then try to break it with an unusual input.

Ask

"How did you test it? What would you improve?"

7 A regular polygon has an angle of 150° inside each corner. How many sides does it have?

Go further

8 How many of the numbers from 1 to 100 are NOT multiples of 2 or of 3?

9 A machine does × 3, then − 8. Mei puts a number in and gets the SAME number out. What is it? What if the machine does × 3, then + 8?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Code the Pattern

1. 28, 33 (add 5)
2. 9 sticks (3, 5, 7, 9). Each new triangle adds 2 sticks, because it shares one side with the triangle before it.
3. 34 (7, 10, 13, 16, 19, 22, 25, 28, 31, 34; the rule $3n + 4$ gives $3 \times 10 + 4 = 34$ too)
4. 24, 20 (take away 4)
5. 705. The rule is $7n + 5$, and $7 \times 100 + 5 = 705$.
6. No. The rule is $7n - 5$. The 29th term is 198 and the 30th term is 205, so 200 is skipped.
7. 19 houses. The rule is $5n + 1$: 19 houses use 96 sticks (4 left over) and 20 houses would need 101.
8. 100 tiles. Pattern 11 has 121 tiles, so 21 more. The differences are 3, 5, 7, 9, ..., so this is NOT a constant-difference sequence.
9. At the 12th term they both say 61. The gap starts at 77 and closes by 7 each time (Ben +5, Mei −2): $77 \div 7 = 11$ jumps after the 1st term. Check: $5 \times 12 + 1 = 61$ and $85 - 2 \times 12 = 61$.

Week 1 • Lesson 2: Prime Hunters

1. 1, 2, 3, 6, 9, 18
2. 21 is not prime (factors 1, 3, 7, 21). 23 is prime (its only factors are 1 and 23).
3. 1 ($4 \times 7 = 28$, so 1 is left over: $29 \bmod 4 = 1$)
4. $30 = 2 \times 3 \times 5$ uses three numbers, and you cannot use four: none of 2, 3 and 5 can be split again. (These are the prime factors of 30.)
5. 7: 1, 2, 4, 8, 16, 32, 64 (8×8 counts once).
6. 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.
7. 6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53.
8. 199. Test 2, 3, 5, 7, 11 and 13 ($14 \times 14 = 196$, $15 \times 15 = 225$): none of them divides 199.
9. Ali: 7×13. Mei: $3 \times 5 \times 7$.

Week 2 • Lesson 3: Below Zero

1. −10, −4, −1, 0, 2
2. 5 °C
3. −5 °C
4. −11 °C, by 5 degrees
5. 2 (work backwards: $-1 - 5 = -6$, then $-6 + 8 = 2$; check: $2 - 8 = -6$, $-6 + 5 = -1$)
6. 23 m (14 up to sea level, then 9 more)
7. Biggest 4 ($5 - 3 - (-2)$ or $5 - (-2) - 3$); smallest −10 ($-2 - 3 - 5$). Try all 6 orders: the answers are 4, 4, 0, 0, −10, −10.
8. One answer: top row −3, 2, 1; middle row 4, 0, −4; bottom row −1, −2, 3. 0 must be in the middle and 4 and −4 cannot be in corners; the other 7 answers are turns and flips of this one.
9. 17 °C ($-6 + 7 \times 2 = 8$ at 7 a.m., then $8 + 3 \times 3 = 17$)

Week 2 • Lesson 4: Algebra Machines

1. 25 ($6 \times 4 = 24$, then + 1)
2. It doubles IN and then adds 1 (OUT = $2n + 1$). IN 10 gives 21.
3. 18
4. 7 ($38 + 4 = 42$, $42 \div 6 = 7$)
5. OUT = $6n - 5$; IN 20 gives 115.
6. $n = 8$ ($65 + 7 = 72$, $72 \div 9 = 8$)
7. Both give 24. $(n + 5) \times 2 = 2n + 10$, so they always give the same OUT.
8. 2 ($2 \times 3 - 4 = 2$; $3n - 4 = n$)
9. 15 (63 → 31 → 15; check 15 → 31 → 63)

Week 3 • Lesson 5: Every Possible Way

1. 15 (3 × 5)
2. 6 numbers: 46, 49, 64, 69, 94, 96
3. 6 (3 + 2 + 1)
4. 24 (4 × 6)
5. 5 (4 × 3 = 12 and 60 ÷ 12 = 5)
6. 24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others)
7. 30 (6 × 5: each ordered pair is one home match)
8. 10 for the pentagon (list them, or 5 × 4 × 3 ÷ 6); 20 for the hexagon (6 × 5 × 4 ÷ 6).
9. 648 (9 choices for the first digit, 9 for the second because 0 is allowed but the first digit is used, 8 for the third: 9 × 9 × 8)

Week 3 • Lesson 6: Fair or Unfair?

1. 3 out of 6 = 1/2 (2, 4 or 6)
2. 10 counters; 7/10
3. About 100 (half of 200), but probably not exactly 100
4. 2/4 = 1/2
5. 4/10 = 2/5 (2, 3, 5, 7)
6. 4/36 = 1/9 (3+6, 4+5, 5+4, 6+3)
7. 15 (5 out of 20 is 1/4)
8. 3/8 (HHT, HTH, THH out of 8 outcomes)
9. 14/30 = 7/15 (10 multiples of 3, 6 of 5, and 15 and 30 are counted twice: 10 + 6 − 2 = 14)

Week 4 • Lesson 7: Turtle Geometry

1. 360° (4 × 90°): one full turn
2. 360° (6 × 60°). The same total as the square: one full turn.
3. 120 (3 × 120 = 360)
4. 5 repeats (5 × 72 = 360). It draws a regular pentagon.
5. 10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°)
6. 1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 − 2) × 180° = 1260°)
7. Last turn 100° (360 − 70 − 100 − 90); inside angle 80° (180 − 100)
8. No: 5 × 108° = 540° is not a whole number of turns. 108 × 10 = 1080 = 3 × 360, so 10 repeats close it: a 10-point star that goes round 3 times (each point 72°).
9. 2: turn 80° (round 2 times) and turn 160° (round 4 times). Turn 120° (3 times round) only draws a triangle 3 times; turns of 200°, 280° … give the same stars turned the other way.

Week 4 • Lesson 8: Secret Codes

1. 3 o'clock (15 − 12 = 3)
2. 8 o'clock (20 − 12 = 8, so 20 mod 12 = 8)
3. 3 (45 = 6 × 7 + 3)
4. Thursday (10 mod 7 = 3; Monday + 3 days)
5. E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE)
6. 2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 − 12 = 2)
7. 23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3)
8. Two letters can give the same code letter: A (0) and N (13) both give 0 = A (13 × 2 = 26). So every code letter could stand for two different letters.
9. GAD (C = 2 gives 6 = G, A = 0 gives 0 = A, B = 1 gives 3 = D). No: all 26 letters get different code letters, because 3 and 26 have no common factor.

Week 5 • Lesson 9: Data Detectives

1. 6 (24 ÷ 4)
2. 6 (24 ÷ 4: the 0 counts as a value)
3. 55
4. 10 kg
5. 160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160)
6. Three answers: 0, 6, 10, 12, 12 or 1, 5, 10, 12, 12 or 2, 4, 10, 12, 12. Median 10 is the middle (3rd) number; the mode 12 must appear twice, so the 4th and 5th numbers are 12; the total is 5 × 8 = 40, so the first two add up to 40 − 10 − 24 = 6. (Using 3 and 3 would make 3 a second mode.)
7. Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30.
8. (a) 12 (5 × 8 = 40, 4 × 7 = 28, 40 − 28 = 12). (b) No. Removing a number above the mean (like 12) lowers the mean, but removing a number below the mean raises it. Example: 4, 6, 8, 10, 12 has mean 8; remove 4 and 6, 8, 10, 12 has mean 9.
9. (a) 50 (10 × 32 = 320, 9 × 30 = 270, 320 − 270 = 50). (b) 33: 270 + 60 = 330 and 330 ÷ 10 = 33.

Week 5 • Lesson 10: Showcase Challenge

1. 62 (the nth term is 3n + 2, so 3 × 20 + 2)
2. 2 × 2 × 3 × 5
3. −8 °C
4. 14
5. Pattern 33 (3n + 1 = 100)
6. 83, 89 and 97
7. 12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12)
8. 33 (multiples of 2: 50, of 3: 33, of both (6): 16; 100 − (50 + 33 − 16) = 33)
9. 4 (4 × 3 − 8 = 4). With + 8 it is −4 (−4 × 3 + 8 = −4).